Iteration #01 - Team Formation and Problem Description

Members: Jay Shah, Shriniketh Mukundan, & Tea Adams

Topic: Predictive Modeling for Injury Prevention in Professional Sports (NBA/NHL)

Objective: Develop a predictive model to assess the likelihood of player injuries in the NHL and NBA, with the goal of informing preventative strategies and optimizing player health through data-driven insights.

Roles & Responsibilities:

Jay Shah - Data Engineer/Database Designer: SQL & database design, ER modeling, normalization

Responsibilities:

Design and implement a relational database for player stats, games, and injuries
Handle ETL processes to integrate multiple data sources and ensure quality

Tea Adams - Data Analyst/Predictive Modeling Lead: Programming, analysis, problem-solving, and predictive modeling

Responsibilities:

Analyze historical data to identify injury risk factors
Build predictive models to assess injury likelihood
Compute derived metrics to support preventative strategies

Shriniketh Mukundan - Data Visualization/Reporting Lead: Data visualization, communication, collaboration

Responsibilities/Roles:

Create dashboards showing injury trends and high-risk players
Present insights clearly through visualizations and reports
Collaborate with the team to interpret results and refine analyses
Reasoning:

Two members of our team are interested in pursuing careers in professional sports post-graduation, while the third is aiming for a career in consulting. We chose a project focused on injury prevention because it allows us to apply data management, predictive analytics, and visualization skills in a real-world sports context. For the two pursuing sports careers, this project provides hands-on experience in an industry we are passionate about and creates a tangible portfolio piece to present to future employers. For the team member interested in consulting, the project develops analytical problem-solving and data-driven decision-making skills that are directly transferable to consulting work. Overall, the project aligns with both our career goals while allowing the team to demonstrate competencies in database design, ETL, predictive modeling, and visualization.